5.20(e)

```r
set.seed(123)

sigma2 <- 1000
N_values <- c(100, 500, 1000, 5000)
R <- 5000   # number of simulations

# function to compute quantities for one simulated sample of x
one_run <- function(N, sigma2 = 1000) {
  # generate x ~ Uniform(0,10)
  x <- runif(N, min = 0, max = 10)

  # sort x in increasing order
  x <- sort(x)

  # split sample into first half and second half
  n1 <- N / 2
  x1 <- x[1:n1]
  x2 <- x[(n1 + 1):N]

  # sample means of two halves
  xbar1 <- mean(x1)
  xbar2 <- mean(x2)

  # sample variance of x using N as divisor
  xbar <- mean(x)
  sx2 <- mean((x - xbar)^2)

  # (i) conditional variances
  # var(b2 | x) = sigma2 / sum((xi - xbar)^2)
  # since sx2 = (1/N) * sum((xi - xbar)^2),
  # sum((xi - xbar)^2) = N * sx2
  var_b2_given_x <- sigma2 / sum((x - xbar)^2)

  # var(beta2_mean | x)
  # = var((ybar2 - ybar1)/(xbar2 - xbar1) | x)
  # = var(ybar2 - ybar1 | x) / (xbar2 - xbar1)^2
  # = [sigma2/(N/2) + sigma2/(N/2)] / (xbar2 - xbar1)^2
  # = (4*sigma2/N) / (xbar2 - xbar1)^2
  var_b2mean_given_x <- (4 * sigma2 / N) / (xbar2 - xbar1)^2

  # (ii) quantities whose expectations are requested
  inv_sx2 <- 1 / sx2
  four_over_gap2 <- 4 / (xbar2 - xbar1)^2

  c(
    var_b2_given_x = var_b2_given_x,
    var_b2mean_given_x = var_b2mean_given_x,
    inv_sx2 = inv_sx2,
    four_over_gap2 = four_over_gap2
  )
}

# run simulations for each N
results_list <- lapply(N_values, function(N) {
  sims <- replicate(R, one_run(N, sigma2))
```

```
  sims <- t(sims)

  data.frame(
    N = N,
    E_var_b2_given_x = mean(sims[, "var_b2_given_x"]),
    E_var_b2mean_given_x = mean(sims[, "var_b2mean_given_x"]),
    E_inv_sx2 = mean(sims[, "inv_sx2"]),
    E_4_over_gap2 = mean(sims[, "four_over_gap2"])
  )
})

results <- do.call(rbind, results_list)

print(results)
```

```
##      N E_var_b2_given_x E_var_b2mean_given_x E_inv_sx2 E_4_over_gap2
## 1  100       1.22102696           1.64733901 0.1221027     0.1647339
## 2  500       0.24102734           0.32215916 0.1205137     0.1610796
## 3 1000       0.12017835           0.16039221 0.1201783     0.1603922
## 4 5000       0.02400743           0.03202002 0.1200371     0.1601001
```